

Effortless Hall Lights Controller



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An assembly hall in an educational institution or industrial shed has multiple light fixtures for lighting purposes. The user has to manipulate a number of switches to switch on a certain number of light fixtures in order to get the desired level of lighting. Also, lights are not switched off when lighting is not required. The circuit presented here allows switching of an optimal number of lighting fixtures with just one knob of a potentiometer.

Circuit and working

Fig. 1 shows the circuit diagram of an effortless hall lights controller. The circuit is built around bar-graph IC LM3914 (IC1), ten BC557 transistors (T1 through T10), ten 12V 1C/O relays and some other components.

The heart of the circuit is an LED bar-graph driver IC LM3914. Controlling signals for the ten relays are generated at pins LED1 through LED10 by

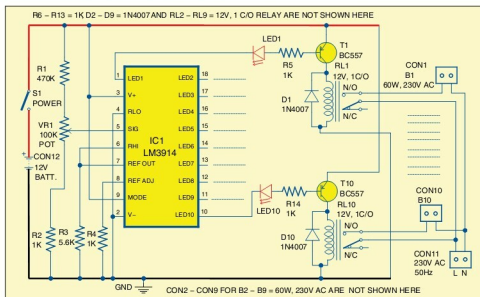


Fig. 1: Circuit diagram of an effortless hall lights controller

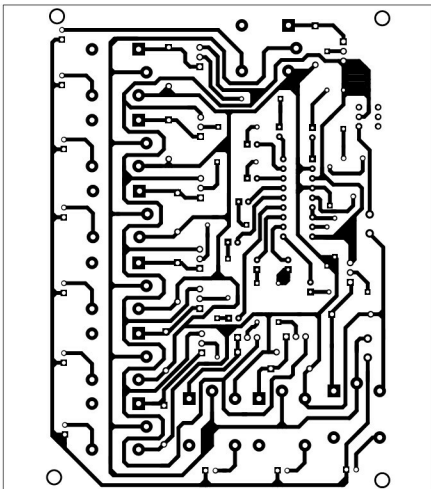


Fig. 2: An actual-size PCB layout of the circuit

PARTS LIST

Semiconductors:

- IC1 - LM3914 bar-graph
- T1-T10 - BC557 pnp transistor
- D1-D10 - 1N4007 rectifier diode
- LED1-LED10 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 470-kilo-ohm
- R2, R4-R14 - 1-kilo-ohm
- R3 - 5.6-kilo-ohm
- VR1 - 100k pot

Miscellaneous:

- CON1-CON10 - 2-pin connector
- CON11, CON12 - 2-pin terminal connector
- RL1-RL10 - 12V, 1C/O relay
- 18-pin IC base
- 12V battery